Received: 5 March 2024 | Revised: 19 November 2024 | Accepted: 27 November 2024

DOI: 10.1002/ejsc.12235

ORIGINAL PAPER

Official ECSS Journal Open Access WILEY

Effect of cold-water immersion treatment on recovery from exercise-induced muscle damage in the hamstring

Yuh-Chuan Huang[1] | Tai-Ying Chou[2] | Trevor C. Chen[3] | Hung-Ting Chen[1]

[1]Physical Education Office, Ming Chuan University, Taipei City, Taiwan

[2]Department of Athletic Performance, National Taiwan Normal University, Taipei City, Taiwan

[3]Department of Physical Education and Sport Sciences, National Taiwan Normal University, Taipei City, Taiwan

Correspondence
Hung-Ting Chen, Physical Education Office, Ming Chuan University, Taipei City, Taiwan.
Email: simonchendr@gmail.com

Abstract

This study investigated the effect of five consecutive days of cold-water immersion (CWI) on recovery from exercise-induced muscle damage (EIMD) in the hamstrings following maximal eccentric contraction (EC) exercise. Eighteen healthy adult women were randomly assigned to a CWI group and a control group (CG) ($n = 9$/group). Participants performed 10 sets of 10 repetitions of isokinetic EC at 30°/second and underwent maximum voluntary isometric contraction (MVC), delayed onset muscle soreness (DOMS) assessment, straight leg raise (SLR) test, and plasma myoglobin (Mb) measurement. The CWI group received one 14-min session of CWI treatment (14°C) at 1, 25, 49, 73, and 97 h after the EC test, whereas the CG rested in a seated position at the same five time points without receiving treatment. (1) All the dependent variables in the CWI group and CG exhibited significant changes after the EC test ($p < 0.05$). (2) The recovery effect in the CWI group was significantly greater than in the CG in terms of the MVC, DOMS, SLR, and plasma Mb concentration results. MVC increased by 89.3 ± 2.0% on the fourth day ($p < 0.013$), DOMS decreased by 15.4 ± 1.5 mm on the second day ($p < 0.000$), SLR increased by 86.3 ± 1.1% on the second day ($p < 0.014$), and plasma Mb decreased by 436.3 ± 60.8% on the third day ($p < 0.014$). The study indicates that five consecutive days of CWI at 14°C significantly enhance recovery from exercise-induced muscle damage in the hamstrings.

KEYWORDS
cryotherapy, delayed onset muscle soreness, maximal voluntary isometric contraction, straight leg raise

Highlights

- CWI treatment was beneficial for recovery from EIMD after high-intensity exercise.
- All the participants exhibited similar and significant responses in their MVC, DOMS, SLR, and plasma Mb concentration results after the EC test.
- The CWI group demonstrated mostly superior recovery responses compared with the CG after receiving the CWI treatment on 5 consecutive days after the EC test.

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2024 The Author(s). European Journal of Sport Science published by Wiley-VCH GmbH on behalf of European College of Sport Science.

1 | INTRODUCTION

Resistance training is often considered an accessible form of physical activity for beginners. However, inexperienced eccentric exercises can lead to exercise-induced muscle damage (EIMD) (Clarkson & Hubal, 2002). Unfamiliar eccentric exercises may cause significant and temporary muscle damage (Huang et al., 2020; Hyldahl & Hubal, 2014; Jamurtas et al., 2013), often accompanied by elevated levels of enzymes such as creatine kinase (CK) and myoglobin (Hyldahl & Hubal, 2014; Peake et al., 2005). Current studies have confirmed that high-intensity physical activity can cause exercise-induced muscle damage (EIMD) and subsequently affect exercise performance (Chen et al., 2011, 2019; Hausswirth et al., 2011; Loturco et al., 2016; Nunes et al., 2020). Chen et al. (2011) observed a significant increase in DOMS and plasma myoglobin (Mb) concentrations after the human knee flexors were subjected to maximum eccentric contraction; by the fifth day after the exercise, the delayed onset muscle soreness (DOMS) had almost disappeared, but the plasma Mb concentration had not decreased. Subsequently, Chen et al. (2019) observed that young men with sedentary lifestyles experienced decreased maximum voluntary isometric contraction (MVC) on the first to fourth days after performing eccentric contraction exercise on the lower limbs but also experienced a significant increase in DOMS and their Mb concentrations. The hamstrings, which are relatively prone to injury, often experience EIMD after high-intensity eccentric contraction training, leading to reduction in MVC and increases in DOMS and plasma Mb concentrations. Introducing proactive and effective interventions into the recovery process can facilitate physiological healing and enhance recovery.

One study demonstrated that CWI at a temperature below 15°C is an effective method for postexercise recovery. This treatment was shown to reduce DOMS compared to a resting control group (CG) (Bleakley et al., 2012). Kwiecien and McHugh (2021) found that cryotherapy at 15°C has been effective for rapid recovery 3–6 h after resistance training, as it helps maintain reduced muscle temperature. Therefore, to minimize injuries following intense exercise and to prevent the spread of tissue damage, cryotherapy should be applied intensively within the first few hours after exercise. Cryotherapy can help lower core and tissue temperatures, alter blood flow, and assist in heat dissipation. Its key benefits include analgesic effects and a reduction in sensory nerve conduction velocity (Allan et al., 2022). The recovery benefits of CWI are hypothesized to stem from its vasoconstrictive effects, which reduce hypoxic cell death or damage, thereby decreasing inflammation and tissue damage (Tipton et al., 2017).

CWI was a popular recovery strategy to assist the recovery of high-intensity exercise. Typical CWI methods include immersing the limbs or trunk in the water temperature of approximately 8–15°C for approximately 5–20 min (Versey et al., 2013). Vaile et al. (2008) discovered leg strength trained males was treated using CWI immediately after, and at 24, 48 and 72 h postexercise, showed that it was beneficial to leg muscle strength recovery. However, some similar studies have found that the application of CWI to resistance training has a negative effect on the increase in muscle strength in male participants (Fyfe et al., 2019; Roberts et al., 2015; Yamane et al., 2015). In a study comparing male and female athletes with EIMD (exercise-induced muscle damage), Delextrat et al. (2013) observed that male and female basketball players who underwent immediate post-competition treatments, including massage, intermittent cold water immersion, or a control condition, showed no significant differences in repeated sprint performance measured 24 h later. However, in terms of recovery, both the massage and cold water immersion groups exhibited lower levels of overall and leg fatigue compared to the control group. Moreover, cold water immersion was found to be more effective than massage in promoting recovery, with the effects being particularly pronounced in females. King and Duffield (2009) investigated the effects of cold water immersion (CWI) on muscle damage in female athletes. Ten female netball players participated in a repeated exercise protocol over several days, followed by immediate recovery interventions, including active recovery (ACT), CWI, or contrast water therapy (CTWT). The results revealed that none of the recovery strategies improved subsequent performance during follow-up exercise. However, CWI was more effective than ACT in reducing muscle soreness. Athletes also expressed a preference for CTWT or CWI, as these methods involved less physical exertion. The authors suggested that future research should explore the acute effects of recovery interventions on performance, as well as the impact of more intense exercise conditions.

CWI would affect the recovery from EIMD. However, further research is required to fully address the symptoms of EIMD. Stephens et al. (2017) pointed out that using CWI to promote exercise recovery is variable and not suitable for everyone. It interacted with individual characteristics (such as physique traits), differences in water-immersion protocol (depth, duration, and temperature), and exercise type. Due to the limited number of studies investigating the effects of cold water immersion (CWI) for recovery from muscle damage induced by resistance training in women, the existing literature primarily focuses on athletes, with little attention given to healthy adult women engaging in more intense exercise. CWI has been widely employed and shown to be effective in mitigating fatigue and muscle soreness associated with exercise-induced muscle damage (EIMD). However, most of the research has concentrated on athletic populations. Therefore, the present study aims to address this deficiency by utilizing healthy adult female participants, subjecting them to high-intensity resistance training, and monitoring their recovery over a continuous 5-day period. This approach represents one of the novel contributions of the current research.

In summary, CWI is a cost-effective, easily implementable, and low-risk recovery method. It helps reduce the severity of EIMD after exercise. In addition, it serves as a suitable method for proactive exercise recovery. However, the literature has not yet confirmed the recovery effects of CWI treatment on the human hamstrings after EIMD induced by eccentric contraction exercise. Considering that hamstring injuries are common among athletes and people who exercise regularly, the present study explored the recovery effects of CWI treatment after hamstring injury. The aim of this study is to investigate the effects of CWI treatment on the recovery of the hamstring muscles following EIMD caused by maximal eccentric contraction exercise. Subsequently, the study will observe whether five consecutive days of

15367290, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ejsc.12235 by Ohio State University Ohio Sta, Wiley Online Library on [18/12/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

CWI treatment can promote the recovery of the hamstring muscles following EIMD, reduce the severity of muscle damage, and accelerate the recovery of exercise performance.

2 | METHODOLOGY

2.1 | Research participants

Eighteen healthy adult women (age: 21.7 ± 1.1 years; height: 162.5 ± 0.8 cm; weight: 54.8 ± 1.0 kg; body fat: 26.1 ± 0.5%) who voluntarily provided consent to participate in this study were recruited. These women were asked to complete a health questionnaire survey and sign a health survey form to confirm their current health status. In addition, they were required to have no history of muscular or skeletal injuries within the 6 months preceding the commencement of this study and to have no medical conditions that could hinder their participation. Before the experiment, we informed the participants of the research purpose and content, the requirements for cooperation, and the potential risks. After each of the participants had signed the informed consent form, the experiment commenced. The participants were instructed to (1) refrain from all types of training and recovery treatment (e.g., massage, infrared radiation, stretching, and icing) from 1 week before the commencement of the experiment to the end of the experiment; (2) refrain from taking painkillers, anti-inflammatory drugs, or nutritional supplements from 2 days before the commencement of the experiment to the end of the experiment; and (3) refrain from consuming stimulants (e.g., caffeine) and alcohol from 1 day before the commencement of the experiment to the end of the experiment. The experimental environment was maintained at 25°C, and all the experiment sessions were scheduled between 1 and 5 p.m.

2.2 | Experimental procedures

On the day before the formal experiment, each participant underwent a series of pretests related to body composition, MVC, DOMS, straight leg raise (SLR), and plasma Mb concentration. The participants were then randomly divided into a CWI group ($n = 9$) and a CG ($n = 9$) on the basis of a double-blind experimental design and the principle of balanced sequencing. No significant differences were observed between the baseline characteristics of the two groups ($p > 0.05$). On the day of the experiment, both the CWI group and the CG performed a single round of 10 sets × 10 repetitions of maximal isokinetic eccentric contraction (30°/second) with the nondominant leg positioned atop an isokinetic dynamometer (Biodex System 3 Pro; NY, USA). Tests related to MVC, DOMS, SLR, and plasma Mb concentration were performed before the eccentric contraction test and at 0.5, 24, 48, 72, 96, and 120 h after the eccentric contraction test. The CWI group received a 14-min session of CWI treatment (at 14°C) at 1, 25, 49, 73, and 97 h after the eccentric contraction test, whereas the CG rested in a seated position at the same five time points without receiving treatment (Table 1). In addition, to ensure the participants safety, body temperature changes during CWI were monitored throughout the experiment. Temperature measurements were taken every 5 min, from pre-immersion to 30 min post-immersion, using an infrared forehead thermometer (FORA IR42 Infrared Thermometer) by the researcher. If the temperature dropped below 36°C, measurements were discontinued.

2.3 | Cold water immersion

Machado et al. (2017) confirmed that cold water immersion at 14°C enhances recovery from exercise-induced muscle damage (EIMD) caused by eccentric contractions. In contrast, Fuchs et al. (2020) showed that leg CWI at 8°C for 20 min after resistance exercise reduced myofibrillar protein synthesis, potentially hindering muscle conditioning. Based on these findings, it is recommended to avoid CWI at temperatures below 14°C and durations longer than 14 min. In the present study, EIMD was induced in the hamstrings by eccentric contraction exercise and was subsequently treated using CWI with the following settings: water temperature of 14°C, seated position, and duration of 14 min. The CWI treatment process is described as follows: Within 5 min after eccentric contraction exercise, each participant was seated in a hydrotherapy pool (water temperature: 14 ± 1°C) for 14 min. The immersion depth was adjusted to reach the anterior superior iliac spine, as suggested by Bouzid et al. (2018). To maintain the water temperature at 14 ± 1°C, we checked this temperature every 3 min by using a thermometer. If any increase in the water temperature was detected, an appropriate amount of ice would be added to the pool, which was then gently stirred to maintain the temperature to below 15°C. During the experiment, the room temperature was maintained at 25 ± 1°C, and the relative humidity was maintained at 50%–60%.

2.4 | Eccentric contraction test

A Biodex machine was employed to perform eccentric contraction on the nondominant leg of each participant. During training, each participant lay prostrate on the Biodex machine with the ankle of the leg being tested secured to the support frame and with the other leg and waist fixed in position with straps. The length of the lever arm was adjusted to fit the position of the leg being tested to ensure that the participant could exert the full strength of their hamstring and to prevent the involvement of other body parts from affecting the results. Subsequently, each participant performed one round of 10 sets × 10 repetitions (100 repetitions in total) of eccentric contraction with their hamstring. The angular velocity was set at 30°/second, and the range of motion was set to be full extension (knee flexion at 90° to complete extension at 0°). Each

15367290, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ejsc.12235 by Ohio State University Ohio Sta, Wiley Online Library on [18/12/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 Description of the experimental protocols.

Day	Item	Experimental protocols
1	Pretest	Each participant underwent a series of pretests related to body composition, MVC, DOMS, SLR, and Mb
2	EC + test CWI & CG	Each participant performed 10 sets of 10 eccentric knee flexion (EC). + Test (MVC, DOMS, SLR, and Mb)
		CWI = received a 14-min session of CWI treatment (at 14°C). CG rested in a seated position at the same time
3–6	Test CWI & CG	Both the CWI group and the CG tests related to MVC, DOMS, SLR, and Mb were performed after the eccentric contraction test. The CWI group received a 14-min session of CWI treatment (at 14°C)
		The CG rested in a seated position at the same time
7	Posttest	Each participant underwent a series of pretests related to MVC, DOMS, SLR, and Mb

Abbreviations: CG, control group, CWI, cold-water immersion, DOMS, delayed onset muscle soreness, EC, eccentric contraction, Mb, plasma myoglobin, MVC, maximum voluntary isometric contraction, SLR, straight leg raise test.

complete repetition was followed by a 10-s rest period, and each set was followed by a 2-minute rest period.

2.5 | Maximum voluntary isometric contraction tes

Each participant performed a 30° MVC test with the hamstrings of their nondominant leg on the Biodex machine. The test was conducted during the pretest and from before the eccentric contraction test to the fifth day after the eccentric contraction test. During the MVC test, the participant lay prostrate on the Biodex machine with their uninvolved leg and waist fixed in position with straps to prevent the involvement of other body parts from affecting the results. The knee joint of the leg being tested was set at a 30° angle. Each knee flexion movement lasted for 5 s and was followed by a 45-second rest period. The test was conducted three times, and the average of the measured values was recorded as the MVC result.

2.6 | Dependent variables

2.6.1 | Delayed onset muscle soreness assessment

The muscle soreness assessment was conducted using a visual analog scale to evaluate the level of muscle soreness in the hamstrings of the nondominant leg after the eccentric contraction test (Chen et al., 2011). The muscle soreness assessment was performed during the pretest and from before the eccentric contraction test to the fifth day after the eccentric contraction test. During the muscle soreness assessment, the participant lay prostrate on a massage table, while a researcher, who was trained in controlled pressure application, used the index and middle fingers of the hand used for stabilization to apply pressure to the muscle belly (the most prominent muscular location) of the participant's rectus femoris. The participant recorded their perceived pain level on a 100-mm visual analog scale with 0 mm indicating no pain and 100 mm indicating severe pain.

2.6.2 | Straight leg raise test

SLR is the most commonly used method for assessing the flexibility of the hamstrings and can be performed passively or actively (Medeiros et al., 2016). Flexibility can be defined as the ability of muscles (or muscle groups) to lengthen and is affected by muscle, connective tissue, and neural tissues (Medeiros et al., 2019). In the present study, we conducted a passive SLR test. This test was conducted during the pretest and from before the eccentric contraction test to the fifth day after the eccentric contraction test. During the SLR test, the participant lay prostrate on a massage table with their head and body in neutral positions, with their arms placed at their sides, and with the backs of their hands facing downward. During the test, their hip joint remained stable, their knee joint was kept straight at 180°, and their ankles remained relaxed. An electronic goniometer (Klein, IP42 Digital Electronic) was placed 15 cm above the lateral ankle joint of the leg being tested. When the test began, the participant was asked to slowly raise their leg while keeping it straight until they were unable to raise it farther, at which point they were to say "stop" so that the angle could be recorded. The test was conducted three times, and the average of the measured values was recorded as the SLR result. During the test, each participant was reminded to stay as relaxed as possible when raising their leg to prevent minor involuntary muscle movements in the hamstrings from affecting the result (Foo et al., 2019).

2.6.3 | Plasma myoglobin measurement

Plasma Mb sampling was conducted during the pretest and from before the eccentric contraction test to the fifth day after the eccentric contraction test. A blood sample of 1 mL was collected from each participant's cubital vein and placed into a vacutainer containing

15367290, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ejsc.12235 by Ohio State University Ohio Sta, Wiley Online Library on [18/12/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

ethylenediaminetetraacetic acid (Becton Dickinson and Company, Plymouth, UK). Subsequently, the blood sample was centrifuged at 3000 rpm for 10 min, and the plasma was then extracted and stored at −80°C in a freezer. Finally, the plasma Mb concentration was analyzed using a fully automated clinical chemistry analyzer (Model Elecsys 2010, Roche Diagnostics GmbH, Mannheim, Germany) along with test reagents (Roche Diagnostics, Indianapolis, IN, USA).

2.7 | Statistical analysis

The obtained values (MVC, DOMS, SLR, and plasma Mb concentrations) were subjected to a one-way analysis of variance (ANOVA). Independent sample *t*-tests were performed to examine the differences in the pretest values of all the dependent variables between the CWI group and the CG. Additionally, a mixed-design two-way ANOVA was conducted to compare the differences in all the dependent variables between the CWI group and the CG at multiple time points. If significant interactions were detected, Scheffé's post hoc test was also performed. Statistical significance was set at $\alpha = 0.05$.

3 | RESULTS

As shown in Figure 1, both the CWI group and the CG exhibited similar responses in MVC, DOMS, SLR, and plasma Mb concentration after maximal isokinetic EC with the hamstrings ($p > 0.05$). However, after the eccentric contraction test, the results for MCV, DOMS, SLR, and plasma Mb concentration in the CWI group at some of the analyzed time points indicated significantly superior recovery compared with the corresponding results in the CG ($p < 0.05$). The monitoring results of body temperature changes obtained every 5 min throughout the CWI treatment revealed no intergroup differences ($p > 0.05$).

3.1 | Pretest

The pretest results of all the dependent variables (i.e., MVC, DOMS, SLR, and plasma Mb concentration) in the CWI group and CG are presented as follows: The MVC results for the nondominant-leg hamstrings were 53.4 ± 1.4 and 52.6 ± 2.2 Nm, respectively; the DOMS results were both 0 mm; the plasma Mb concentrations were 12.5 ± 0.4 and 12.6 ± 1.6 μg/L, respectively; the SLR results were 92.1% ± 2.0% and 91.4% ± 1.5%, respectively. No significant differences were observed in any of the dependent variables between the CWI group and CG ($p > 0.05$).

3.2 | MVC test

On the first day to the third day after the eccentric contraction test, no significant differences were observed in the MVC results for the nondominant-leg hamstrings between the CWI group and CG

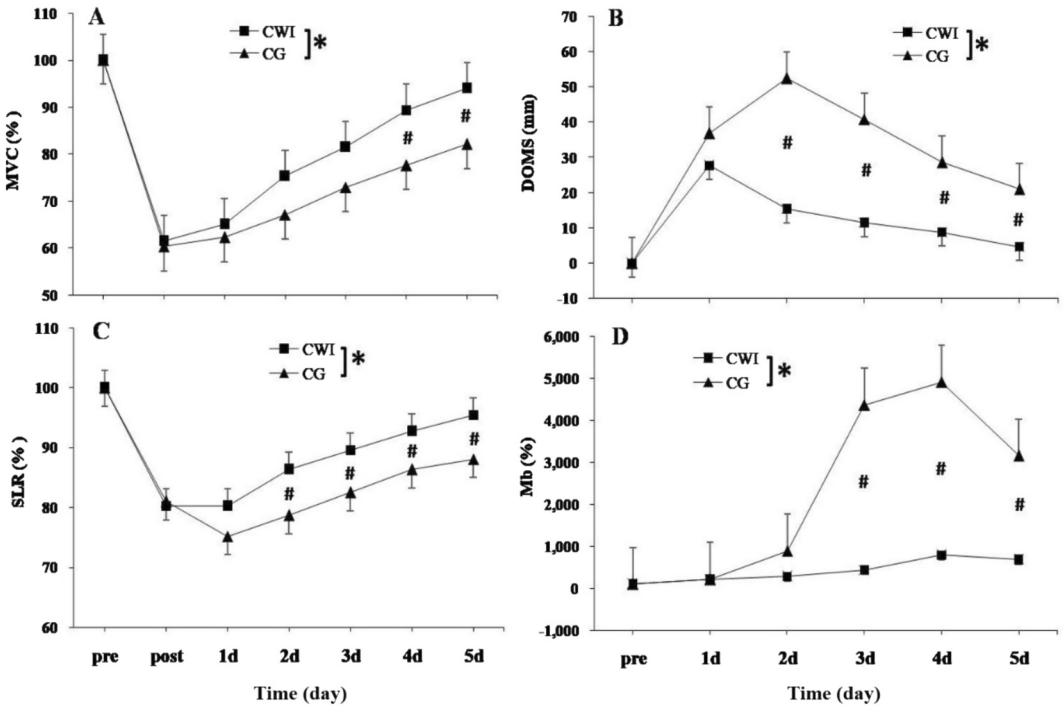

FIGURE 1 Changes in the results of maximum voluntary isometric contraction (MVC), delayed onset muscle soreness (DOMS), straight leg raise (SLR), and plasma myoglobin (Mb) concentration at multiple time points after the participants received the cold-water immersion treatment (CWI). In panels (A)–(D), * indicates the MVC, DOMS, SLR, and plasma Mb concentration results of the CWI group and the control group (CG) revealed significant interaction effects ($p < 0.05$). # indicates significant differences between the MVC, DOMS, SLR, and plasma Mb concentration results of the CWI group and CG at multiple time points ($p < 0.05$).

15367290, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ejsc.12235 by Ohio State University Ohio Sta, Wiley Online Library on [18/12/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

($p > 0.05$; Figure 1A). However, the MVC results of the CWI group on the fourth day (89.3% ± 2.0%) and fifth day (94.0% ± 1.5%) were significantly more favorable than those of the CG (77.6% ± 2.3% and 82.1% ± 2.0% on the fourth and fifth days, respectively) ($p < 0.013$ and $p < 0.005$; Figure 1A).

3.3 | DOMS assessment

The CWI group and CG exhibited significantly similar DOMS responses in the nondominant-leg hamstrings 1 day after the eccentric contraction test ($p < 0.05$; Figure 1B). After receiving five rounds of CWI treatment on five consecutive days after the eccentric contraction test, the CWI group experienced a reduced DOMS response compared with that of the CG, which received no treatment during the 5 days after the eccentric contraction test. From the second day to the fifth day after the eccentric contraction test, the rate of DOMS reduction in the CWI group was significantly higher than that in the CG ($p < 0.000$; Figure 1B). For example, the DOMS results of the CWI group and CG on the second day were 15.4 ± 1.5 and 52.4 ± 1.6 mm, respectively, and those on the third day were 11.6 ± 1.5 and 40.8 ± 1.4 mm, respectively.

3.4 | SLR test

No significant differences in the SLR results for the nondominant-leg hamstrings between the CWI group and CG were observed immediately after the eccentric contraction test or on the first day after the eccentric contraction test ($p > 0.05$; Figure 1C). However, the SLR results of the CWI group on the second day to the fifth day after the eccentric contraction test were significantly more favorable than those of the CG ($p < 0.05$; Figure 1C). For example, the SLR results of the CWI group and CG on the second day after the eccentric contraction test were 86.3% ± 1.1% and 78.8% ± 1.1% ($p < 0.014$), respectively, and those on the third day were 89.6% ± 1.0% and 82.5% ± 0.8% ($p < 0.010$), respectively.

3.5 | Plasma Mb measurement

No significant differences in the plasma Mb concentrations of the CWI group and CG were observed on the first day or second day after the eccentric contraction test ($p > 0.05$; Figure 1D). However, the plasma Mb concentration results of the CWI group on the third day to the fifth day after the eccentric contraction test were 436.3% ± 60.8%, 785.8% ± 100.2%, and 684.4% ± 115.0%, respectively, which demonstrated a significantly greater rate of Mb concentration decline than that of the CG (4362.1% ± 788.5% ($p < 0.014$), 4912.3% ± 708.6% ($p < 0.010$), and 3151.4% ± 324.3% ($p < 0.011$) on the third day to the fifth day after the eccentric contraction test, respectively) ($p < 0.05$; Figure 1D).

4 | DISCUSSION

In this study, we recruited healthy adult women to perform 100 repetitions of maximal eccentric contraction on the hamstrings of their nondominant leg, and we discovered that CWI treatment (water temperature of 14°C, seated position, and duration of 14 min) was beneficial for recovery from EIMD after high-intensity exercise. Our findings are described as follows: (1) All the participants exhibited similar and significant responses in their MVC, DOMS, SLR, and plasma Mb concentration results after the eccentric contraction test. (2) The CWI group demonstrated mostly superior recovery responses compared with the CG after receiving the CWI treatment on 5 consecutive days after the eccentric contraction test ($p < 0.05$). These findings supported the hypothesis of this study by revealing that five sessions of the CWI treatment accelerated the recovery of MVC muscle strength, reduced DOMS, restored hamstrings flexibility, and decreased the plasma Mb concentration (Figure 1A–D, respectively). Given the previous literature on the negative effects of repeated CWI in males, such as attenuated chronic skeletal muscle adaptations from resistance exercise (Hyldahl & Peake, 2020), detrimental impacts on muscle mass and strength gains (Fyfe et al., 2019; Roberts et al., 2015; Yamane et al., 2015), and impaired muscle protein synthesis rates (Fuchs et al., 2020; Roberts et al., 2015). However, this study is limited by the research design and cannot conduct research or discussion on chronic skeletal muscle adaptations and muscle protein synthesis rates issues. It is recommended that these important factors can be added for comparison in future studies.

All the participants performed 10 sets × 10 repetitions (100 repetitions in total) of maximal isokinetic eccentric contraction (30°/second) with their nondominant-leg hamstrings; this exercise resulted in a notable decrease in MVC muscle strength, with the effect lasting for 5 days (Figure 1A). This finding was consistent with that of Chen et al. (2019), namely that young men with sedentary lifestyles experienced a decline in MVC muscle strength in the first 4 days after performing eccentric contraction on their hamstrings. The CWI intervention in the present study demonstrated a positive recovery effect on the hamstrings following a decline in MVC muscle strength caused by eccentric contraction. Furthermore, on the fourth and fifth days, the recovery effect in the CWI group was significantly superior to that in the CG, indicating that the CWI treatment was beneficial to the restoration of muscle strength in the hamstrings of healthy women after high-intensity exercise. This finding was consistent with the viewpoint of Huang et al. (2020) in that both investigated maximal eccentric contraction training for the hamstrings, showing reduced maximal voluntary contraction (MVC) and increased muscle soreness. However, Vaile et al. (2008) found that male participants who received whole-body cold water immersion (CWI) at 15°C for 14 min after leg resistance training had improved squat MVC and reduced soreness, while our study observed significant MVC improvements in the CWI group only on the fourth and fifth days. This discrepancy may be due to differences in exercise intensity,

15367290, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ejsc.12235 by Ohio State University Ohio Sta, Wiley Online Library on [18/12/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

immersion depth, and gender. Our research on females shows that five sessions of 14°C CWI (14 min each) over 5 days effectively enhance MVC recovery. Future studies should explore whether varying immersion depths could further improve recovery. This study found that CWI at 14°C for 14 min is beneficial for the recovery of MVC in female participants after intense resistance exercise. This result aligns with similar studies indicating that CWI negatively impacts muscle strength gains in male participants following resistance training (Fyfe et al., 2019; Roberts et al., 2015; Yamane et al., 2015). The observed differences may be related to gender, as well as variations in CWI temperature and duration. For example, Fyfe et al. (2019) used CWI at 10°C for 15 min, Roberts et al. (2015) at 10°C for 10 min, and Yamane et al. (2015) at 10°C for 20 min. In contrast, the CWI temperature in this study was 14°C, which is closer to the 15°C used by Kwiecien and McHugh (2021), who found that this temperature effectively supports rapid recovery 3–6 h post-resistance training. Additionally, Fuchs et al. (2020) demonstrated that leg CWI at 8°C for 20 min following single resistance-type exercises in healthy men reduced myofibrillar protein synthesis rates, potentially impairing muscle conditioning. Therefore, it is recommended to avoid CWI at temperatures lower than 14°C and durations exceeding 14 min.

In the present study, in the first 2 days after the eccentric contraction test, no significant differences were observed between the MVC results of the CWI group and those of the CG. Furthermore, the observation period in the present study was 5 days, and five CWI treatment sessions were held; this procedure may have been a crucial factor contributing to the significant recovery of MVC muscle strength in the CWI group (approaching the pretest levels) on the fourth and fifth days after the eccentric contraction test. Accordingly, we recommend that healthy adult populations that emphasize the recovery of MVC muscle strength in the hamstrings should consider employing CWI treatment to expedite the recovery of their MVC capacity for subsequent high-intensity training sessions.

The DOMS changes in the nondominant-leg hamstrings of all the participants 1 day after the eccentric contraction test were significant compared with the DOMS level before the eccentric contraction test ($p < 0.05$; Figure 1B), and the pain associated with DOMS persisted for 5 days before recovery to the pretest levels. In addition, the present study observed that the CWI treatment (14°C seated CWI for 14 min for 2–5 consecutive days) greatly accelerated the alleviation of DOMS. The CWI group exhibited milder DOMS responses after five CWI treatment sessions following the eccentric contraction test compared with the CG, which received no recovery treatment. Moreover, notably, the alleviation speed increased from the second day to the fifth day. This result aligned with that of Leeder et al. (2012) and Vanderlei et al. (2017), who revealed that CWI is beneficial to the reduction of postexercise muscle soreness. Delextrat et al. (2013) found that cold water immersion (CWI) reduced fatigue more effectively than massage in female basketball players. King and Duffield (2009) showed that CWI did not improve performance in subsequent exercise for female netball players. Machado et al. (2017) induced EIMD in participants through eccentric contraction and discovered that their CWI groups (CWI1: 9°C for 15 min; CWI2: 14°C for 15 min) exhibited superior muscle pain recovery compared with their CG. In addition, the recovery of CWI2 was significantly faster than that of CWI1. It is evident that a water temperature of 14°C is effective in promoting the recovery of DOMS, and there is no necessity to use lower temperatures, such as 9°C, as this may result in excessive discomfort for the individuals submerged. Furthermore, the CWI treatment in that study was confirmed to have no harmful effects. Accordingly, intervention with CWI treatment (9–14°C for 5–15 min) after high-intensity exercise reduces pain, particularly at a water temperature of 14°C and an immersion duration of 15 min. Therefore, CWI is a safe and effective recovery method for healthy men and women seeking to reduce muscle pain after high-intensity exercise.

All the present participants' SLR flexibility in the nondominant-leg hamstrings immediately after the eccentric contraction test was significantly lower than the pre-exercise level (Figure 1C). This finding aligned with that of Clarkson and Hubal (2002), who revealed that the symptoms of EIMD include pain, stiffness, and swelling. Furthermore, no significant differences were observed between the SLR results of the CWI group and CG immediately after the eccentric contraction test or 1 day after the eccentric contraction test ($p > 0.05$). However, the recovery effect in the CWI group was significantly greater than that in the CG from the second day to the fifth day ($p < 0.05$). These findings indicate that the hamstrings experienced pain, stiffness, and swelling after overload eccentric contraction, which affected its ability for at least five days and may have impacted its subsequent athletic performance. Future studies could further explore this aspect. Medeiros et al. (2016, 2019) stated that the SLR test is commonly used to assess the flexibility of the hamstrings (biceps femoris, semitendinosus, and semimembranosus). Flexibility is affected by muscle, connective tissue, and neural tissues and is considered to be a contributing factor to hamstring strains recurring with high frequency. In the present study, the CWI group exhibited faster recovery in hamstrings flexibility after the eccentric contraction than did the CG, indicating that CWI treatment is suitable as an active recovery method following high-intensity exercise.

In the literature, the recovery effects of CWI treatment on the blood plasma Mb concentration in the hamstrings following EIMD induced by eccentric contraction remain unclear. However, in the present study, we observed that the peak concentration of plasma Mb in the CG occurred on the fourth day after the eccentric contraction test, constituting a rapid increase in this concentration compared with that in the CWI group (Figure 1D). This finding was consistent with that of Chen et al. (2011, 2019), who discovered on multiple occasions that the plasma Mb concentration in the hamstrings significantly increased after maximal eccentric contraction. We observed that the plasma Mb concentration in the CWI group experienced no surge between the first day and the fifth day after the eccentric contraction test, suggesting that the CWI treatment sessions at 1, 25, 49, 73, and 97 h after the eccentric contraction test mitigated the speed of the increase in the plasma Mb concentration. This effect may have been due to the effect of CWI on reducing

15367290, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ejsc.12235 by Ohio State University Ohio Sta, Wiley Online Library on [18/12/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

tissue temperature, metabolic rate, and inflammation response, as proposed by Brukner and Khan (1993), and also the capability of CWI to prevent hypoxic cell death and damage and further minimize the inflammation response and tissue damage, as explained by Tipton et al. (2017). These findings indicate that the hamstrings are prone to EIMD after high-intensity exercise, leading to a rapid increase in its plasma Mb concentration. CWI treatment over multiple consecutive days can accelerate the reduction of the plasma Mb concentration. These findings are similar to those of Higgins et al. (2017), Machado et al. (2017), and Vanderlei et al. (2017), all of whom have suggested that CWI treatment is harmless to the human body and is effective in expediting recovery from EIMD in both generally healthy individuals and athletes who participate in team sports.

5 | CONCLUSION

The study demonstrates that five consecutive days of cold water immersion at 14°C significantly enhances recovery from exercise-induced muscle damage in the hamstrings. Participants in the CWI group showed greater improvements in MVC, DOMS, SLR, and plasma Mb concentration compared to the control group. These findings suggest that CWI is an effective recovery strategy for accelerating muscle recovery and mitigating muscle damage after intense exercise.

ACKNOWLEDGEMENTS

We would like to thank all the participants of this study for their contribution, which made the completion of the research possible.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

ORCID

Hung-Ting Chen https://orcid.org/0000-0001-9439-0924

REFERENCES

Allan, R., J. Malone, J. Alexander, S. Vorajee, M. Ihsan, W. Gregson, S. Kwiecien, and C. Mawhinney. 2022. "Cold for Centuries: A Brief History of Cryotherapies to Improve Health, Injury and Post-Exercise Recovery." *European Journal of Applied Physiology* 122(5): 1153–62. https://doi.org/10.1007/s00421-022-04915-5.

Bleakley, C., S. McDonough, E. Gardner, G. D. Baxter, J. T. Hopkins, and G. W. Davison. 2012. "Cold-Water Immersion (Cryotherapy) for Preventing and Treating Muscle Soreness After Exercise." *Cochrane Database of Systematic Reviews* 2012(2): CD008262. https://doi.org/10.1002/14651858.cd008262.pub2.

Bouzid, M. A., K. Ghattassi, W. Daab, S. Zarzissi, M. Bouchiba, L. Masmoudi, and H. Chtourou. 2018. "Faster Physical Performance Recovery With Cold Water Immersion Is Not Related to Lower Muscle Damage Level in Professional Soccer Players." *Journal of Thermal Biology* 78: 184–91. https://doi.org/10.1016/j.jtherbio.2018.10.001.

Brukner, P., and K. Khan. 1993. *Clinical Sports Medicine*. McGraw-Hill Book: Company Australia Pty.

Chen, T. C., K.-Y. Lin, H.-L. Chen, M.-J. Lin, and K. Nosaka. 2011. "Comparison in Eccentric Exercise-Induced Muscle Damage Among Four Limb Muscles." *European Journal of Applied Physiology* 111(2): 211–23. https://doi.org/10.1007/s00421-010-1648-7.

Chen, T. C., T.-J. Yang, M.-J. Huang, H.-S. Wang, K.-W. Tseng, H.-L. Chen, and Kazunori Nosaka. 2019. "Damage and the Repeated Bout Effect of Arm, Leg and Trunk Muscles Induced by Eccentric Resistance Exercises." *Scandinavian Journal of Medicine & Science in Sports* 29(5): 725–35. https://doi.org/10.1111/sms.13388.

Clarkson, P. M., and M. J. Hubal. 2002. "Exercise-Induced Muscle Damage in Humans." *American Journal of Physical Medicine & Rehabilitation* 81(11): S52–69. https://doi.org/10.1097/00002060-200211001-00007.

Delextrat A., Calleja-González J., Hippocrate A., & Clarke N. D. (2013). Effects of Sports Massage and Intermittent Cold-Water Immersion on Recovery From Matches by Basketball Players. *Journal of Sports Sciences*, 31(1), 11–9. https://doi.org/10.1080/02640414.2012.719241.

Foo, Y., M. E. Héroux, L. Chia, and J. Diong. 2019. "Involuntary Hamstring Muscle Activity Reduces Passive Hip Range of Motion During the Straight Leg Raise Test: A Stimulation Study in Healthy People." *BMC Musculoskeletal Disorders* 20(1): 130. https://doi.org/10.1186/s12891-019-2511-6.

Fuchs, C. J., I. W. K. Kouw, T. A. Churchward-Venne, J. S. J. Smeets, J. M. Senden, W. D. V. M. Lichtenbelt, L. B. Verdijk, and L. J. C. van Loon. 2020. "Postexercise Cooling Impairs Muscle Protein Synthesis Rates in Recreational Athletes." *The Journal of physiology* 598(4): 755–72. https://doi.org/10.1113/JP278996.

Fyfe, J. J., J. R. Broatch, A. J. Trewin, E. D. Hanson, C. K. Argus, A. P. Garnham, S. L. Halson, R. C. Polman, D. J. Bishop, and A. C. Petersen. 2019. "Cold Water Immersion Attenuates Anabolic Signaling and Skeletal Muscle Fiber Hypertrophy, but Not Strength Gain, Following Whole-Body Resistance Training." *Journal of Applied Physiology* 127(5): 1403–18. https://doi.org/10.1152/japplphysiol.00127.2019.

Hausswirth, C., J. Louis, F. Bieuzen, H. Pournot, J. Fournier, J.-R. Filliard, and J. Brisswalter. 2011. "Effects of Whole-Body Cryotherapy vs. Far-Infrared vs. Passive Modalities on Recovery from Exercise-Induced Muscle Damage in Highly-Trained Runners." *PLoS One* 6(12): 27749. https://doi.org/10.1371/journal.pone.0027749.

Higgins, T. R., D. A. Greene, and M. K. Baker. 2017. "Effects of Cold Water Immersion and Contrast Water Therapy for Recovery from Team Sport: A Systematic Review and Meta-Analysis." *The Journal of Strength & Conditioning Research* 31(5): 1443–60. https://doi.org/10.1519/JSC.0000000000001559.

Huang, T. H., Lin, J. C., Ma, M. C., Yu, T., & Chen, T. C. (2020). "Acute Responses of Bone Specific and Related Markers to Maximal Eccentric Exercise of the Knee Extensors and Flexors in Young Men." *Journal of Musculoskeletal and Neuronal Interactions* 20(2), 206–15.

Hyldahl, R. D., and M. J. Hubal. 2014. "Lengthening Our Perspective: Morphological, Cellular, and Molecular Responses to Eccentric Exercise." *Muscle & Nerve* 49(2): 155–70. https://doi.org/10.1002/mus.24077.

Hyldahl, R. D., and J. M. Peake. 2020. "Combining Cooling or Heating Applications With Exercise Training to Enhance Performance and Muscle Adaptations." *Journal of Applied Physiology* 129(2): 353–65. https://doi.org/10.1152/japplphysiol.00322.2020.

Jamurtas, A. Z., A. Garyfallopoulou, A. A. Theodorou, A. Zalavras, V. Paschalis, C. K. Deli, M. G. Nikolaidis, I. G. Fatouros, and Y. Koutedakis. 2013. "A Single Bout of Downhill Running Transiently Increases HOMA-IR without Altering Adipokine Response in Healthy Adult Women." *European Journal of Applied Physiology* 113(12): 2925–32. https://doi.org/10.1007/s00421-013-2717-5.

King, M., and Rob Duffield. 2009. "The Effects of Recovery Interventions on Consecutive Days of Intermittent Sprint Exercise." *The Journal of Strength & Conditioning Research* 23(6): 1795–802. https://doi.org/10.1519/JSC.0b013e3181b3f81f.

15367290, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ejsc.12235 by Ohio State University Ohio Sta, Wiley Online Library on [18/12/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Kwiecien, S. Y., and M. P. McHugh. 2021. "The Cold Truth: The Role of Cryotherapy in the Treatment of Injury and Recovery from Exercise." *European Journal of Applied Physiology* 121(8): 2125–42. https://doi.org/10.1007/s00421-021-04683-8.

Leeder, J., C. Gissane, K. van Someren, W. Gregson, and G. Howatson. 2012. "Cold Water Immersion and Recovery From Strenuous Exercise: A Meta-Analysis." *British Journal of Sports Medicine* 46(4): 233–40. https://doi.org/10.1136/bjsports-2011-090061.

Loturco, I., C. Abad, F. Nakamura, S. Ramos, R. Kobal, S. Gil, L. Pereira, et al. 2016. "Effects of Far Infrared Rays Emitting Clothing on Recovery After an Intense Plyometric Exercise Bout Applied to Elite Soccer Players: A Randomized Double-Blind Placebo-Controlled Trial." *Biology of Sport* 33(3): 277–83. https://doi.org/10.5604/20831862.1208479.

Machado, A. F., A. C. Almeida, J. K. Micheletti, F. M. Vanderlei, M. F. Tribst, J. Netto Junior, and C. M. Pastre. 2017. "Dosages of Cold-Water Immersion Post Exercise on Functional and Clinical Responses: A Randomized Controlled Trial." *Scandinavian Journal of Medicine & Science in Sports* 27(11): 1356–63. https://doi.org/10.1111/sms.12734.

Medeiros, D. M., A. Cini, G. Sbruzzi, and C. S. Lima. 2016. "Influence of Static Stretching on Hamstring Flexibility in Healthy Young Adults: Systematic Review and Meta-Analysis." *Physiotherapy Theory and Practice* 32(6): 438–45. https://doi.org/10.1080/09593985.2016.1204401.

Medeiros, D. M., L. L. P. Miranda, V. B. Marques, J. B. de Araujo Ribeiro-Alvares, and B. M. Baroni. 2019. "Accuracy of the Functional Movement Soccer (FMS^{TM}) Active Straight Leg Raise Test to Evaluate Hamstring Flexibility in Soccer Players." *International Journal of Sports Physical Therapy* 14(6): 877–84. https://doi.org/10.26603/ijspt20190877.

Nunes, R. F. H., F. J. Cidral-Filho, L. J. F. Flores, F. Y. Nakamura, H. F. M. Rodriguez, F. Bobinski, A. De Sousa, et al. 2020. "Effects of Far-Infrared Emitting Ceramic Materials on Recovery During 2-week Preseason of Elite Futsal Players." *The Journal of Strength & Conditioning Research* 34(1): 235–48. https://doi.org/10.1519/JSC.0000000000002733.

Peake, J. M., K. Suzuki, G. Wilson, M. Hordern, K. Nosaka, L. Mackinnon, and J. S. Coombes. 2005. "Exercise-induced Muscle Damage, Plasma Cytokines, and Markers of Neutrophil Activation." *Medicine & Science in Sports & Exercise* 37(5): 737–45. https://doi.org/10.1249/01.mss.0000161804.05399.3b.

Roberts, L. A., M. Muthalib, J. Stanley, G. Lichtwark, K. Nosaka, J. S. Coombes, and J. M. Peake. 2015. "Effects of Cold Water Immersion and Active Recovery on Hemodynamics and Recovery of Muscle Strength Following Resistance Exercise." *American Journal of Physiology - Regulatory, Integrative and Comparative Physiology* 309(4): R389–98. https://doi.org/10.1152/ajpregu.00151.2015.

Stephens, J. M., S. Halson, J. Miller, G. J. Slater, and C. D. Askew. 2017. "Cold- Water Immersion for Athletic Recovery: One Size Does Not Fit All." *International Journal of Sports Physiology and Performance* 12(1): 2–9. https://doi.org/10.1123/ijspp.2016-0095.

Tipton, M. J., N. Collier, H. Massey, J. Corbett, and M. Harper. 2017. "Cold Water Immersion: Kill or Cure?" *Experimental Physiology* 102(11): 1335–55. https://doi.org/10.1113/EP086283.

Vaile, J., S. Halson, N. Gill, and B. Dawson. 2008. "Effect of Hydrotherapy on the Signs and Symptoms of Delayed Onset Muscle Soreness." *European Journal of Applied Physiology* 102(4): 447–55. https://doi.org/10.1007/s00421-007-0605-6.

Vanderlei, F. M., M C. de Albuquerque, A. C. de Almeida, A. F. Machado, J. Netto Jr, and C. M. Pastre. 2017. "Post-Exercise Recovery of Biological, Clinical and Metabolic Variables after Different Temperatures and Durations of Cold Water Immersion: A Randomized Clinical Trial." *The Journal of Sports Medicine and Physical Fitness* 57(10): 1267–75. https://doi.org/10.23736/s0022-4707.17.06841-4.

Versey, N. G., S. L. Halson, and B. T. Dawson. 2013. "Water Immersion Recovery for Athletes: Effect on Exercise Performance and Practical Recommendations." *Sports Medicine* 43(11): 1101–30. https://doi.org/10.1007/s40279-013-0063-8.

Yamane, M., N. Ohnishi, and T. Matsumoto. 2015. "Does Regular Post-Exercise Cold Application Attenuate Trained Muscle Adaptation?" *International Journal of Sports Medicine* 36(8): 647–53. https://doi.org/10.1055/s-0034-1398652.

15367290, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ejsc.12235 by Ohio State University Ohio Sta, Wiley Online Library on [18/12/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License